MATHS MANTHAN
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Subject: Mathematics | **PERIODIC TEST – 04 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. The ratio of total surface area of a solid hemisphere to the square of its radius is: **[1]**

(a) (a) $2\pi : 1$ (b) $3\pi : 1$ (c) $4\pi : 1$ (d) $1 : 3\pi$

2. If the volume of a sphere of radius R is 16 times the volume of a hemisphere of radius r, then the ratio $\frac{R}{r}$ is: **[1]**

(a) (a) 1 (b) 2 (c) 3 (d) 4

3. If the diagonal of a cube is 17.32 cm, its volume is (Take $\sqrt{3} = 1.732$): **[1]**

(a) (a) 1000 cm^3 (b) 1732 cm^3 (c) 173.2 cm^3 (d) 10000 cm^3

4. The surface area of a cube is 216 cm^2. Its volume is: **[1]**

(a) (a) 144 cm^3 (b) 196 cm^3 (c) 212 cm^3 (d) 216 cm^3

5. **Assertion (A):** Two cubes each of volume 125 cm^3 are joined end-to-end to form a cuboid. The surface area of the resulting cuboid is 250 cm^2.
Reason (R): If two identical cubes of edge a are joined end-to-end, the total surface area of the resulting cuboid is $10a^2$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. If the volumes of two cones are in the ratio 1 : 4 and their diameters are in the ratio 4 : 5, write the ratio of their heights. **[2]**

7. A solid sphere and a solid cube have equal surface areas. What is the ratio of the volume of the sphere to that of the cube? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. The interior of a building is in the form of a cylinder of diameter 4.3 m and height 3.8 m, surmounted by a cone whose vertical height is 2.1 m. Find the total interior surface area and volume of the building. **[3]**

9. A wooden toy rocket is in the shape of a cone mounted on a cylinder. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. Base diameter of cone is 5 cm, while base diameter of cylinder is 3 cm. If the cone is painted orange and cylinder yellow, find the area painted with each colour. (Use $\pi = 3.14$) **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm, are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped into the vessel. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. **Case Study: Adventure Camp Conical Tents and Refreshments**
Students at an adventure camp in Sakleshpur were served juice upon arrival: some in a cylindrical glass ($r = 3.5$ cm, $h = 10.5$ cm) and some in a hemispherical cup ($r = 3.5$ cm). Later, each group of four students was given 551 m^2 canvas to erect a conical tent of base radius 7 m, with 1 m^2 wasted during stitching.

(i) Find the capacity of the cylindrical glass.
(ii) Find the capacity of the hemispherical cup.
(iii) Find the height of the conical tent constructed with the available canvas.
(iv) How much floor area is occupied by each student inside the tent?

[4]

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